Dmitry Lvov

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Research Profile

Medical AI researcher with an honours master's degree from ITMO. My work spans human attention, radiology-report generation and clinical model evaluation. I develop research ideas and methods, design and run experiments, write manuscripts and guide students through their research.

Experience

Innopolis University | AI in Medicine Lab Oct 2023 - present

Researcher, Medical AI

- Design experiments and develop models for medical imaging, human attention and clinical ML evaluation; work with radiologists to define tasks and evaluation protocols and review failure cases.
- **Student supervision:** help students define research problems and protocols, review their code and experiments, check results and revise manuscripts.
- Originated and supervised two NeurIPS 2026 TAE workshop studies, with students as first authors and myself as second author. Wrote the EHR paper and validated and extended its experiments; designed the prenatal experiments and revised the manuscript.
- Originated **SCFR** and led its research and manuscript; co-developed **LogitGaze-Med**, linking medical VLM semantics with radiologists' attention and diagnostic learning. First-author NeurIPS 2026 and 2025 papers.

ITMO University | Mind Lab Dec 2024 - present

Engineer, part-time

- Built multimodal document processing for technical QA: element detection, figure-caption association and vision-language descriptions. This work formed my master's thesis and a first-author IEEE Access article.

Selected Publications

NeurIPS 2026, main conference: Source-Causal Control of Historical Context in Longitudinal Radiology Report Generation. First author.

NeurIPS 2025, main conference: From Human Attention to Diagnosis: Semantic Patch-Level Integration of Vision-Language Models in Medical Imaging. First author.

NeurIPS 2026, TAE workshop: Similar Scores, Different High-Risk Episodes: Auditing Model Selection in Longitudinal EHRs. Second author.

NeurIPS 2026, TAE workshop: Same Top Fraction, Different Workload: Auditing Eligibility and Capacity in Clinical Risk Evaluation. Second author.

ICLR 2025, workshop: LogitGaze: Predicting Human Attention Using Semantic Information from Vision-Language Models. First author.

TAE: Can We Trust AI Evaluation? ICLR workshop: Reasoning and Planning for Large Language Models.

Additional IEEE Access work: multimodal document QA (2026); uncertainty in chest X-ray analysis (2025); cardiometric coefficients (2025).

Education

ITMO University | Master's degree with Honours 2024 - 2026

Applied Mathematics and Computer Science; Big Data and Machine Learning. **ITMO Best Graduate 2026.**

Kazan Federal University | Bachelor's degree in Applied Mathematics 2020 - 2024

Skills and Activities

Python, PyTorch, Transformers, scikit-learn, LangGraph, LangChain, Docker, ClearML, Git and Linux.

English: **IELTS Academic 7.5** (Nov 2025). Track and field: university team member; competes regularly.